



CANDIDATE
NAME

--

CIVICS
GROUP

2

0

-

REGISTRATION
NUMBER

--	--

H1 Biology

8876/02

Paper 2

15 September 2021

Structured Questions and Free Response Question

2 hours

Additional Materials: 12-page Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Candidates are to answer:

Section A on the Question Paper.

Section B on separate 12-page Answer Booklet.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7 OR 8	
Total	60

This document consists of **16** printed pages and **2** blank pages.

BLANK PAGE

Answer **all** questions.

- 1** Hormones are responsible for regulating the permeability of kidney cells to water and dissolved ions.

Kidney cells in the collecting duct are impermeable to water. To allow water molecules to move into cells, the hormone vasopressin upregulates the expression of aquaporin channel proteins.

Fig. 1.1 shows the cell surface membrane of a kidney cell, with five proteins labelled **A** to **E**.

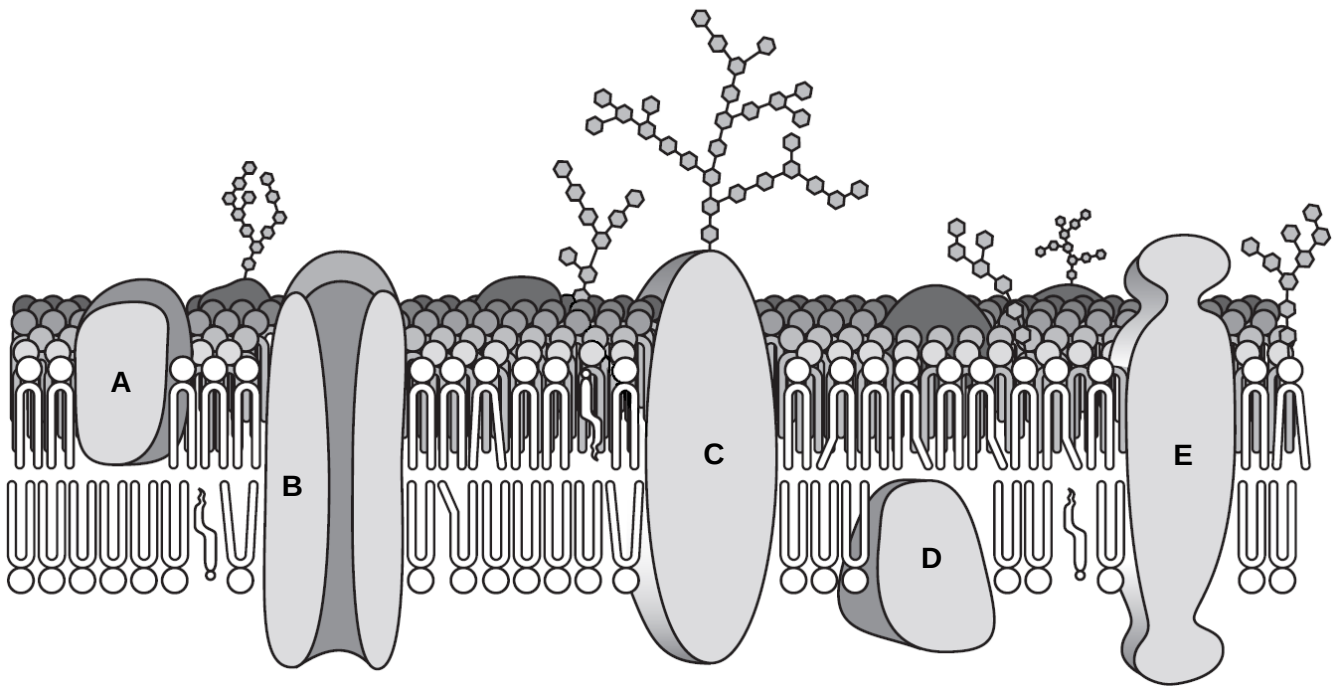


Fig. 1.1

(a) From Fig. 1.1,

- (i)** identify which protein represents the structure of an aquaporin.

.....[1]

- (ii)** draw an arrow on Fig. 1.1 to show the direction of movement of water molecules. [1]

- (iii)** explain how the arrangement of molecules supports the fluid mosaic model.

.....

[2]

Vasopressin receptors are G-protein coupled receptors with a transmembrane domain.

(b) Explain how vasopressin receptors are held in the membrane.

.....

.....

.....

..... [2]

The lipid-soluble hormone aldosterone regulates the level of sodium and potassium ions. It achieves this by upregulating the expression of $\text{Na}^+\text{-K}^+$ pumps. Fig. 1.2 shows how these protein pumps facilitate the transport of sodium ions into the cell and potassium ions out of the cell.

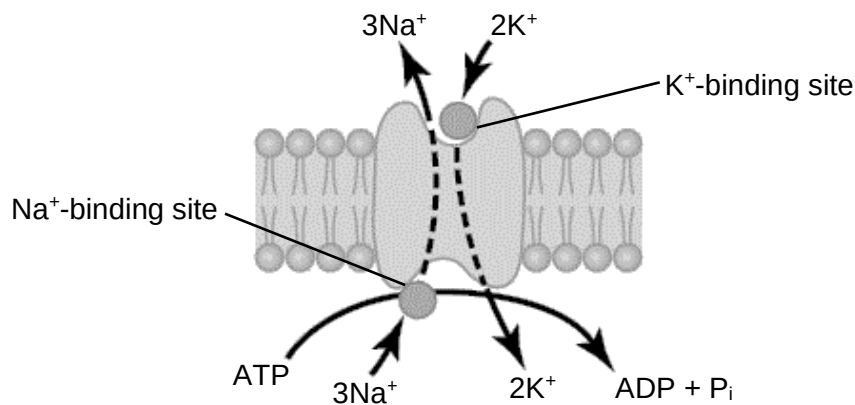


Fig. 1.2

(c) Explain why $\text{Na}^+\text{-K}^+$ pumps allow Na^+ and K^+ to move across the cell surface membrane.

.....

.....

.....

..... [2]

(d) Describe **one** difference in the transport of sodium ions and water molecules into kidney cells in response to aldosterone and vasopressin respectively.

.....

..... [1]

[Total: 9]

- 2 Tuberculosis (TB) is an infectious disease caused by the pathogenic bacterium *Mycobacterium tuberculosis*. In this bacterial species, an enzyme known as fructose 1,6-bisphosphate aldolase (FBA) is critical for glucose metabolism.

Fig. 2.1 shows the relative activity of *M. tuberculosis* FBA measured at different pH levels.

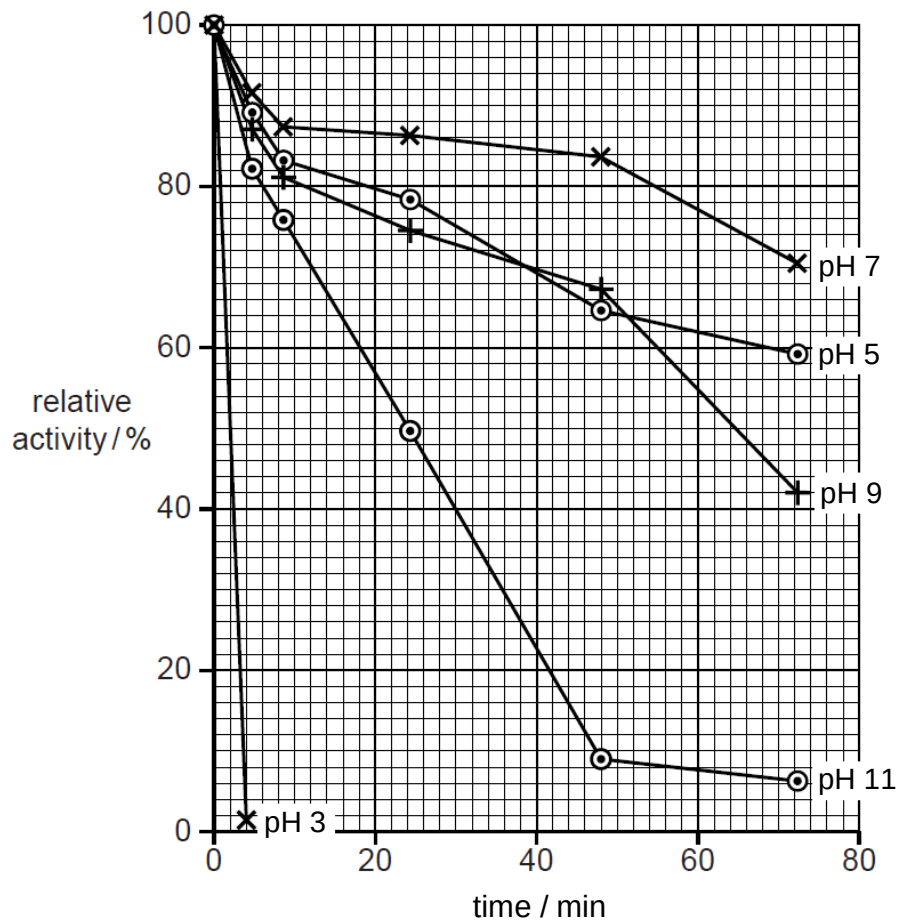


Fig. 2.1

- (a) With reference to Fig. 2.1, distinguish between the effect of neutral and alkaline pH on the activity of FBA.

.....

..... [1]

Fig. 2.2 shows the structure of FBA, which is made of four identical polypeptide subunits.

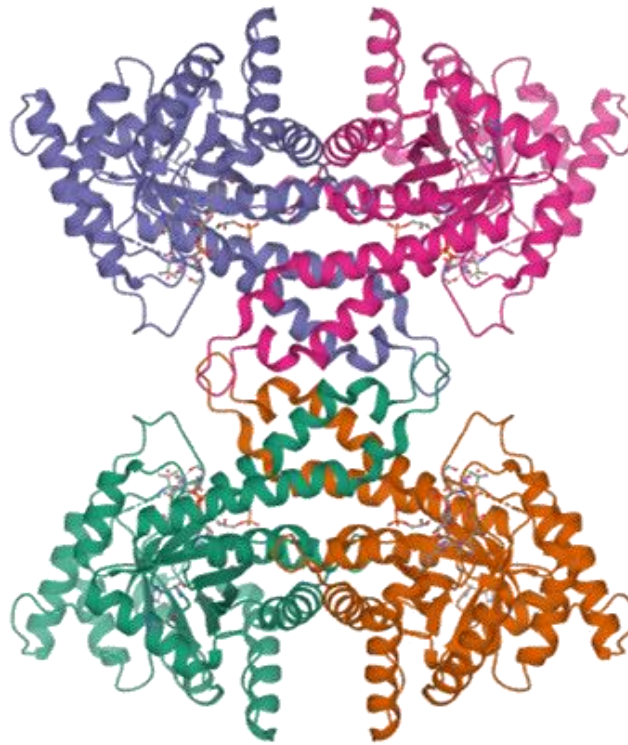


Fig. 2.2

(b) (i) On Fig. 2.2, label and name a secondary structure.

[1]

(ii) Explain how changing the pH level affects the structure and activity of FBA.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

[Total: 6]

3 Fig. 3.1 shows the normal differentiation pathway of a spermatogonial stem cell into spermatids.

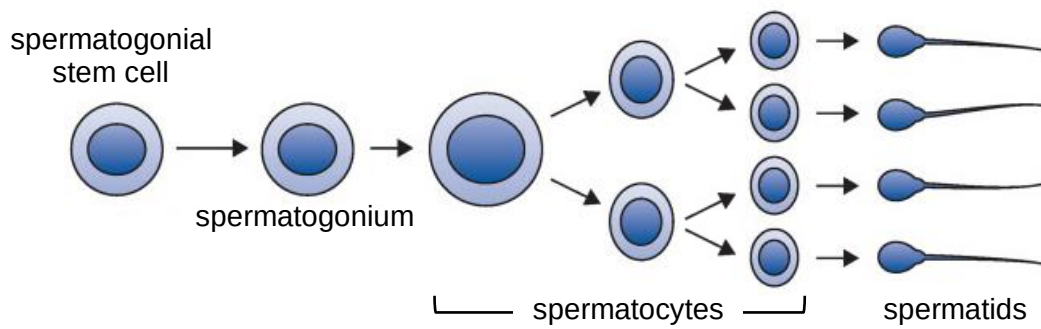


Fig. 3.1

Fig. 3.2 shows three groups of spermatocytes, **A**, **B** and **C**, in different stages of cell division. The peak relative mass of DNA is indicated for each group of cells.

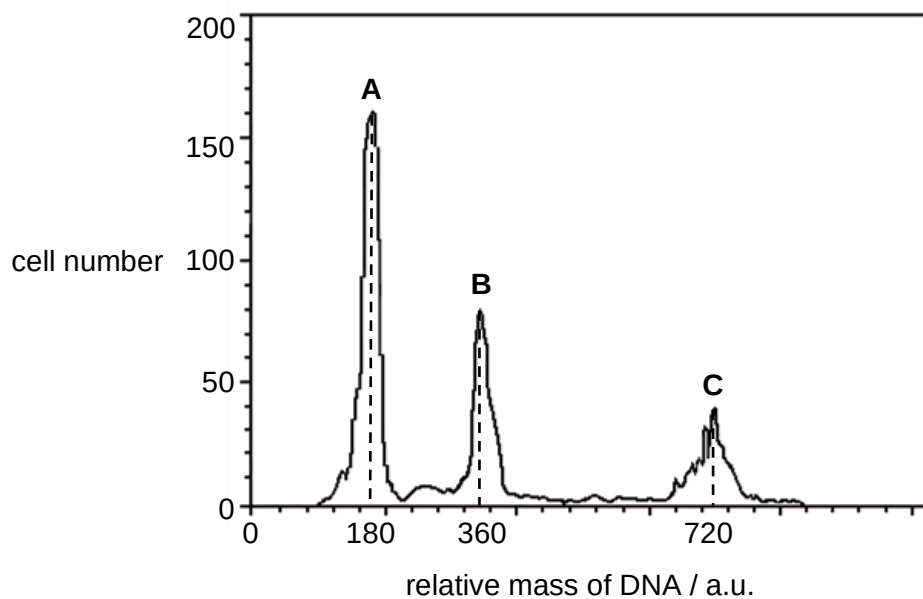


Fig. 3.2

(a) With reference to Fig. 3.2,

(i) identify the group of spermatocytes that are formed at the end of

[1]

meiosis I	meiosis II

(ii) explain your answer in (a)(i).

.....

.....

.....

.....

.....[3]

Germ cell tumours result from uncontrolled division of germ cells, which give rise to gametes. In males between the ages of 15 to 35 years, the risk of testicular cancer increases with age.

Fig. 3.3 shows how a tumour may result from uncontrolled cell division of a spermatogonium (male germ cell), instead of normal differentiation into spermatids.

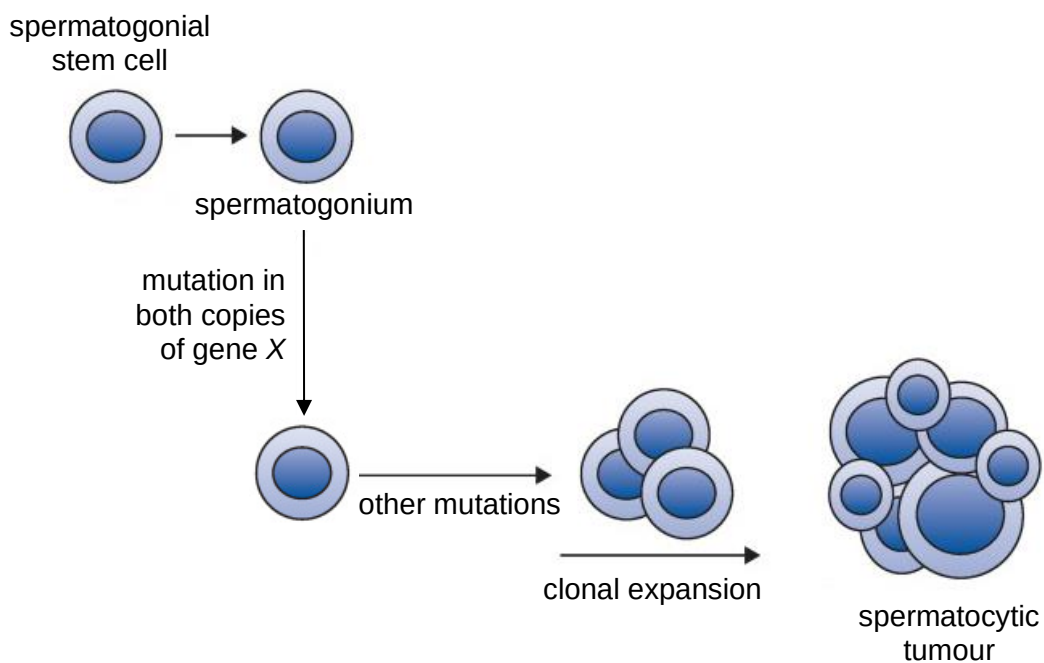


Fig. 3.3

(b) (i) Suggest the name of gene X.

.....[1]

(ii) Explain how gene *X* may result in uncontrolled cell division of a spermatogonium.

.....

.....

.....

.....

.....

..... [3]

(c) Cancer treatments, including chemotherapy and radiotherapy, can often result in loss of fertility. In male cancer patients, sperm cryopreservation (freezing of sperm cells) is one method of preserving fertility.

Using your knowledge of stem cells, suggest why fully differentiated sperm cells are preserved instead of spermatogonial stem cells.

.....

..... [1]

[Total: 9]

- 4 In the bacterium *Thermus thermophilus*, a sequence of DNA contains two overlapping genes, *rnpA* and *rpmH*, which code for RNase P and a ribosomal protein respectively.

Fig. 4.1 shows the first 30 bases in the DNA sequence. Following process X, a different strand of nucleic acid is formed, and it contains the translation start sites for RNase P and the ribosomal protein.

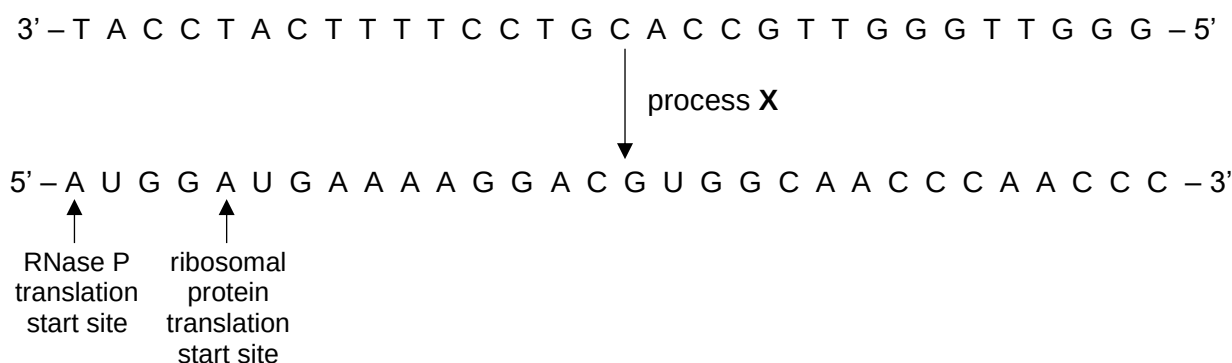


Fig. 4.1

- (a) (i) Briefly describe process X.

.....

.....

.....

.....

.....

.....[3]

- (ii) Suggest why the product of process X can code for two different polypeptide products.

.....

.....

.....

.....[2]

In eukaryotic cells, the *FECH* gene codes for ferrochetalase, which is the final enzyme in the metabolic pathway of haem synthesis.

A study done in zebrafish found that a point mutation in the *FECH* gene resulted in a disease phenotype known as dracula. The disease is characterised by lysis of red blood cells upon exposure to light.

Fig. 4.2 shows a segment of the *FECH* template DNA sequence in normal and dracula zebrafish, and Table 4.1 shows the genetic code.

normal

27	28	29	30	31	32	33
CAC	GTG	GAC	AGT	GGA	CTC	ATC

dracula

27	28	29	30	31	32	33
CAC	GTG	GAC	ATT	GGA	CTC	ATC

Fig. 4.2

Table 4.1

	U		C		A		G		
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC		UCC		UAC		UGC		C
	UUA	Leu	UCA		UAA	STOP	UGA	STOP	A
	UUG		UCG		UAG	STOP	UGG	Trp	G
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC		CCC		CAC		CGC		C
	CUA		CCA		CAA	Gln	CGA		A
	CUG		CCG		CAG		CGG		G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC		ACC		AAC		AGC		C
	AUA	Met	ACA		AAA	Lys	AGA	Arg	A
	AUG		ACG		AAG		AGG		G
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC		GCC		GAC		GGC		C
	GUA		GCA		GAA	Glu	GGA		A
	GUG		GCG		GAG		GGG		G

(b) Using Fig. 4.2 and Table 4.1,

(i) State the amino acid sequence for this segment of the normal polypeptide.

.....[1]

(ii) Explain how the point mutation results in non-functional ferrochetalase.

.....
.....
.....
.....[2]

[Total: 8]

- 5 Marfan syndrome is a genetic disorder that affects the connective tissue. Fig. 5.1 shows the inheritance pattern of Marfan syndrome in a family.

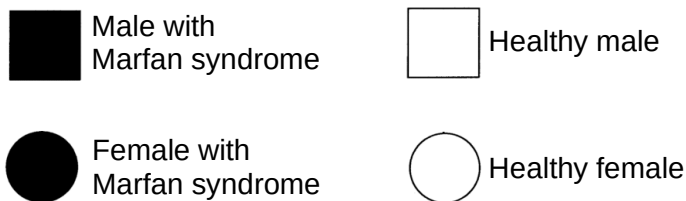
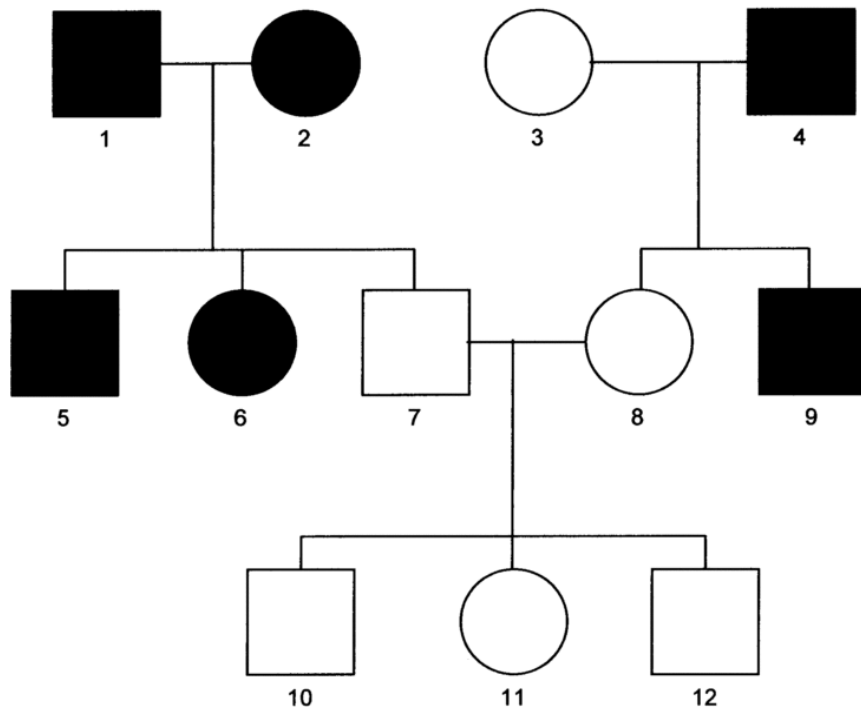


Fig. 5.1

- (a) (i) State if Marfan syndrome is caused by a recessive or dominant allele.

.....[1]

- (ii) Explain your answer to (a)(i).

.....

[2]

- (b) Other than the inheritance of Marfan syndrome, the inheritance of colour blindness was also tracked for the same family. For instance, individuals 3 and 4 have normal vision but their son (individual 9) has colour blindness. Colour blindness is a sex-linked trait.

Based on the mode of inheritance of Marfan syndrome and colour blindness, draw a genetic diagram to show the expected offspring phenotypic ratio that could result for the children born to individuals 3 and 4.

[4]

[Total: 7]

- 6 The coho salmon, *Oncorhynchus kisutch*, is a species of North American salmon. Adult salmon spend majority of their time in the salt water of the Pacific Ocean until it is time for them to migrate back to the natal stream. The natal stream is where the salmon first hatched from eggs, and they return to the natal stream as adults to spawn (i.e. release sperm or eggs for mating).

At the natal stream, adult male salmon engage in physical fights to gain access to female, with larger sized males often emerging as winners and earning the opportunity to mate with females. Some adult males, however, are much smaller in size. These males are known as 'jacks'. Jacks hide around smaller boulders in the streambed to avoid fighting with bigger males. As part of his mating strategy, the jack "sneaks" in close to a female to release his sperms just as the female is depositing her eggs. This increases his chance to fertilize the females' eggs without needing to fight with other bigger males. In this way, both jacks and normal breeding adult males are able to spawn successfully.

Fig. 6.1 shows the variation in body length of reproductively mature males in an original population of coho salmon, before the evolution of the jack phenotype in the population.

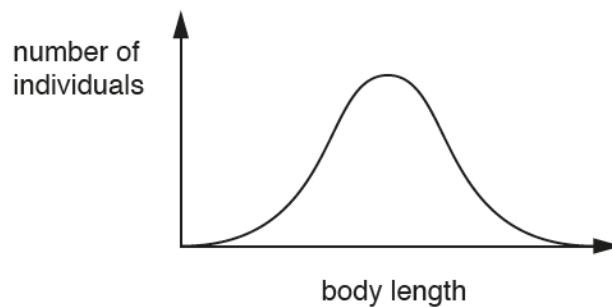


Fig. 6.1

Selection has since acted over time on this original population to change the distribution of body length in reproductively mature male salmon.

- (a) (i) Sketch on Fig. 6.1 to show the new distribution of body length in the present-day population of reproductively mature male salmon. [1]

- (ii) Name the type of selection that is occurring.

.....[1]

In Chilean salmon, the bacteria *Piscirickettsia salmonis* causes the disease piscirickettsiosis, which can have a major impact on farmed salmon populations. To combat this, the antibiotic oxytetracycline is most frequently used to target *P. salmonis* outbreaks.

Table 6.1 shows the percentage of oxytetracycline resistant *P. salmonis* strains in a population of farmed Chilean salmon. The percentage of antibiotic resistant bacterial strains was assessed at 1, 2, 12 and 18 months after the start of oxytetracycline therapy.

Table 6.1

time after start of therapy / months	antibiotic resistant bacterial strains / %
1	2
2	10
12	20
18	50

- (b) With reference to Table 6.1, explain how the use of oxytetracycline therapy can lead to the evolution of antibiotic resistant bacterial strains.

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 6]

Section B

Answer **one** question in this section.

Write your answers on the 12-page answer booklet provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts (a), (b) etc., as indicated in the question.

Either

7 (a) Explain the importance of non-covalent bonds in cellular processes. [9]

(b) Herbicides work in a number of different ways. The herbicide atrazine works by disabling plastoquinone, which is one of the electron carriers in the electron transport chain between photosystem I and II.

Explain how atrazine would kill a susceptible plant. [6]

OR

8 (a) Explain the importance of cyclical processes that occur in a plant. [9]

(b) Water can fill air spaces in the soil surrounding the roots. This prevents oxygen from reaching root hair cells.

Using your knowledge of aerobic and anaerobic respiration, explain why overwatering can kill plants. [6]

BLANK PAGE